University of South Carolina
College of Engineering & Computing
Department of Mechanical Engineering

EMCH380 – Project Management

Homework 1

Due date: Tuesday, September 21 (by 2:49pm)

Notes:
- *Homework should be submitted on Blackboard, hard copies will not be accepted.*
- *SafeAssign will be used to identify areas of overlap between the submitted assignment and existing works.*

1. The widening of a road section requires relocating ("reconductoring") 1700 feet of 13.8-kV overhead primary line. The following table summarizes the activities of the project.

   Construct the associated project network.

Notation	Activity Description	Predecessor(s)	Duration (days)
A	Job review	_	1
B	Advise customers of temporary outage	A	1/2
C	Requisition stores	A	1
D	Scout job	A	1/2
E	Secure poles and material	C,D	3
F	Distribute poles	E	3 1/2
G	Pole location coordination	D	1/2
H	Re-stake	G	1/2
I	Dig Holes	H	3
J	Frame and set poles	F,I	4
K	Cover old conductors	F,I	1
L	Pull new conductors	J,K	2
M	Install remaining material	L	2
N	Sag conductor	L	2
O	Trim trees	D	2
P	De-energize and switch lines	B,M,N,O	1/9
Q	Energize and switch new line	P	1/2
R	Clean up	Q	1
S	Return material to stores	I	2

(25 points)

2. The footings of a building can be completed in four connective sections. The activities for each section include (1) digging, (2) placing steel, and (3) pouring concrete. The digging of one section cannot start until that of the preceding section has been completed. The same restriction applies to pouring concrete. Develop the project network using an Activity on Node representation. *(25 points)*

3. In Problem 2, suppose that after digging all sections, plumbing work can be started but only 20% of the job can be completed before any concrete is poured. After each section of the footings is completed, the remaining plumbing work can be started provided that the preceding 20% portion is complete. Construct the project network using an Activity on Node representation. *(25 points)*

4. Construct an AOA and AON network for a project comprised of activities A to P that satisfies the following precedence relationships: *(25 points)*
   a. A, B, and C, the first activities of the project, can be executed concurrently.
   b. D, E, and F follow A.
   c. I and G follow both B and D.
   d. H follows both C and G.
   e. K and L follow I.
   f. J succeeds both E and H.
   g. M and N succeed F, but cannot start until both E and H are completed.
   h. O succeeds M and I.
   i. P succeeds J, L, and O.
   j. K, N, and P are the terminal activities of the project.